

Lab 1 :

The screenshot shows the DBeaver IDE interface with a PostgreSQL database named 'dvdrental' selected. The query editor displays a SQL query that selects object names, data sizes, index sizes, and total sizes for various tables. The results are shown in a table with 16 rows.

```

SELECT
    fed.relname AS object_name,
    pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
    pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
    pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
FROM pg_class AS fed
JOIN pg_namespace AS n
    ON n.oid = fed.relnamespace
WHERE fed.relkind = 'r'
    AND n.nspname = 'public'
ORDER BY pg_total_relation_size(fed.oid) DESC;

```

	object_name	data_size	index_size	total_size
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	film_category	72 kB	40 kB	112 kB
9	city	64 kB	48 kB	112 kB

Total rows: 16 Query complete 00:00:00.380

Lab 2:

The screenshot shows the DBeaver IDE interface with a PostgreSQL database named 'dvdrental' selected. The query editor displays a SQL query that selects object names, data sizes, index sizes, and total sizes for various tables. The results are shown in a table with 16 rows.

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    fed.relname AS object_name,
    pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
    pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
    pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
FROM pg_class AS fed
JOIN pg_namespace AS n
    ON n.oid = fed.relnamespace
WHERE fed.relkind = 'r'
    AND n.nspname = 'public'
ORDER BY pg_total_relation_size(fed.oid) DESC;
SELECT pg_size_pretty ( pg_indexes_size ( 'payment ' ) ) ;
SELECT pg_size_pretty(pg_indexes_size('payment'));

```

	pg_size_pretty
1	920 kB

Query History

```

442     pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
443 FROM pg_class AS fed
444 JOIN pg_namespace AS n
445     ON n.oid = fed.relnamespace
446 WHERE fed.relkind = 'r'
447     AND n.nspname = 'public'
448 ORDER BY pg_total_relation_size(fed.oid) DESC;
449 SELECT
450     pg_indexes_size('payment') AS after_bytes,
451     pg_size_pretty(pg_indexes_size('payment')) AS after_size;
452 SELECT
453     pg_relation_size('idx_payment_date_long') AS index_bytes,
454     pg_size_pretty(pg_relation_size('idx_payment_date_long')) AS index_size,
455     CEIL(
456         pg_relation_size('idx_payment_date_long')::numeric / 8192

```

Data Output Messages Notifications

Showing rows: 1

	after_bytes bigint	after_size text
1	1523712	1488 kB

dvdrental/postgres@PostgreSQL 18

Query History

```

11 ORDER BY pg_total_relation_size(fed.oid) DESC;
12 SELECT pg_size_pretty ( pg_indexes_size ( 'payment ' ) ) ;
13 SELECT pg_size_pretty(pg_indexes_size('payment'));
14 CREATE INDEX idx_payment_date_long ON payment (payment_date , amount,
15 customer_id ) ;
16 SELECT
17     pg_indexes_size('payment') AS after_bytes,
18     pg_size_pretty(pg_indexes_size('payment')) AS after_size;
19 SELECT
20     pg_relation_size('idx_payment_date_long') AS index_bytes,
21     pg_size_pretty(pg_relation_size('idx_payment_date_long')) AS index_size,
22     CEIL(
23         pg_relation_size('idx_payment_date_long')::numeric / 8192
24     ) AS pages_8kb;
25

```

Data Output Messages Notifications

Showing rows: 1

	index_bytes bigint	index_size text	pages_8kb numeric
1	581632	568 kB	71

Lab 3

Dashboard X Properties X SQL X Statistics X Dependencies X Processes X dvdrental/postgres@PostgreSQL 18* X

dvdrental/postgres@PostgreSQL 18

No limit

Query Query History Scratch Pad

```
1 CREATE INDEX idx_expensive_films
2 ON film (replacement_cost)
3 WHERE replacement_cost > 25.00;
4 SELECT pg_size_pretty(
5     pg_relation_size('idx_expensive_films')
6 );
7 CREATE INDEX idx_full_replacement_cost
8 ON film (replacement_cost);
9 SELECT
10     pg_size_pretty(pg_relation_size('idx_expensive_films')) AS partial_index,
11     pg_size_pretty(pg_relation_size('idx_full_replacement_cost')) AS full_index;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	partial_index text	full_index text
1	16 kB	40 kB